Lab 3: Component Modelling & Architectural Pattern Selection

Micro-Lending & Peer Credit Risk Assessor - Architectural Justification

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Architecture Selection:

We chose **Microservices Architecture** for the Micro-Lending & Peer Credit Risk Assessor System.

Reason 1: Independent Scalability

The platform handles distinct operational domains with varying resource demands. The Risk Assessor component evaluates computationally heavy heuristic rules (such as debt-to-income ratios and employment stability indicators). A microservices design allows this specific engine to scale independently during high-volume loan application periods, ensuring that the system operates efficiently without unnecessarily duplicating UI or payment resources.

Reason 2: Fault Isolation & System Resilience

In a peer-to-peer financial system, mitigating single points of failure is crucial. Microservices offer strong fault isolation, meaning that if one non-critical service fails, the entire application remains functional. For instance, if the background notification or risk scoring service experiences temporary downtime, Lenders and Borrowers can still access the UI and the Loan Management component to view active portfolios or make EMI payments.

Security Advantage: Controlled Access to Financial Profiles

A microservices architecture enforces strict domain-level security and data isolation. The Presentation Layer cannot query the database directly; all interactions must pass through the Loan Manager API gateway. Sensitive financial history and repayment records are protected behind the internal Risk and Payment microservices. This design drastically reduces attack vectors by preventing a compromised frontend from accessing raw database tables or manipulating risk assessment heuristics.

Performance Benefit: Asynchronous Processing & Reduced Latency

Decoupling the components enables asynchronous, event-driven operations. When a borrower processes an EMI repayment, the Payment Service can immediately commit the ACID transaction and then asynchronously notify the Risk Assessor to adjust the credit score. This non-blocking approach ensures the UI remains highly responsive and eliminates transaction bottlenecks, particularly during peak load times.

Components Identified

1. **User Interface Component** - Provides the web/mobile interaction portal for Lenders and Borrowers, requiring the Loan API.
2. **Loan Manager Component** - Orchestrates core lending workflows, providing the Loan API and requiring Risk and Payment interfaces.
3. **Risk Assessor Component** - Processes heuristic rules to compute credit scores, providing the Risk API.
4. **Payment Service Component** - Manages secure, ACID-compliant EMI transactions, providing the Payment API.
5. **Data & Storage Component** - Houses user financial profiles and loan histories, providing the DB API.

